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Imtiaz Ahmad (Albaloshi)



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VU PIONEERS OF ZOOLOGY

1. Immunology is the study of:
 - A. Enzymes
 - B. Hormones
 - C. **Immune system** ✓
 - D. Nervous system
2. The primary function of antibodies is to:
 - A. Catalyze reactions
 - B. Provide energy
 - C. **Identify and neutralize antigens** ✓
 - D. Store genetic information
3. Which of the following is NOT an arm of adaptive immunity?
 - A. Humoral immunity
 - B. Cell-mediated immunity
 - C. **Inflammatory response** ✓
 - D. Immune memory
4. Antigens are:
 - A. **Disease causing agents** ✓
 - B. Antibodies
 - C. WBCs
 - D. Complement proteins
5. The smallest antigenic determinant on an antigen molecule is called:
 - A. **Epitope** ✓
 - B. Paratope
 - C. Hapten
 - D. Idiotope
6. Which cell is responsible for presenting antigen to helper T cells?
 - A. B cell
 - B. **Macrophage** ✓
 - C. Neutrophil
 - D. Plasma cell
7. Immunoglobulin M (IgM) is the first antibody produced in response to:
 - A. Secondary antigen exposure
 - B. **Primary antigen exposure** ✓
 - C. Chronic infection
 - D. Vaccination
8. The main role of Complement system is:
 - A. Antigen recognition
 - B. Clot formation
 - C. **Inflammation and lysis of pathogens** ✓
 - D. Antibody production
9. Which of the following is an example of innate immunity?
 - A. Production of antibodies
 - B. Immune memory
 - C. **Phagocytosis** ✓
 - D. Clonal selection
10. IL-2 (Interleukin-2) is mainly produced by:
 - A. B cells
 - B. Macrophages
 - C. **Helper T cells** ✓
 - D. Plasma cells
11. Which immunoglobulin is most abundant in secretions like saliva, tears and breast milk?
 - A. **IgA** ✓
 - B. IgD
 - C. IgE
 - D. IgG
12. Which cells are responsible for cell-mediated immunity?
 - A. B lymphocytes
 - B. Plasma cells
 - C. **T lymphocytes** ✓
 - D. Eosinophils
13. The process in which B cells produce antibodies is called:
 - A. Clonal deletion
 - B. **Clonal expansion** ✓
 - C. Phagocytosis
 - D. Chemotaxis
14. Which cytokine activates macrophages?
 - A. IL-4
 - B. **IL-12** ✓
 - C. IL-10
 - D. TGF-β
15. Hypersensitivity type I is also known as:
 - A. Cytotoxic hypersensitivity
 - B. Delayed-type hypersensitivity
 - C. **Anaphylactic hypersensitivity** ✓
 - D. Arthus reaction
16. Which cells release histamine in allergic reactions?
 - A. Macrophages
 - B. **Basophils** ✓
 - C. Neutrophils
 - D. Epithelial cells
17. The tolerance to self-antigens is established mainly in:
 - A. **Bone marrow and thymus** ✓
 - B. Spleen and lymph nodes
 - C. Liver and kidney
 - D. Skin and lungs
18. Which molecule is responsible for self-nonself discrimination in vertebrates?
 - A. Immunoglobulin
 - B. **MHC molecules** ✓
 - C. Complement C3
 - D. Cytokines
19. Autoimmunity occurs when:
 - A. Immune system is suppressed
 - B. **Immune system reacts against self antigens** ✓
 - C. Antigens are not recognized
 - D. Antibodies are not produced
20. A vaccine works by:
 - A. Causing disease
 - B. Suppressing immunity
 - C. **Providing active immunity** ✓
 - D. Destroying antigens



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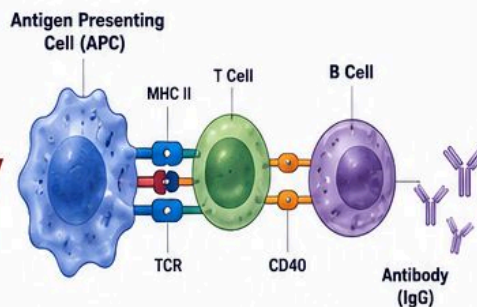
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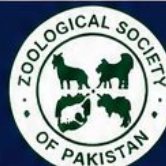


BT302 IMMUNOLOGY

SET 2



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- Which immunoglobulin is mainly responsible for mucosal immunity?
 - IgG
 - IgM
 - IgA ✓**
 - IgD
- The hypervariable region of an antibody is called:
 - Constant region
 - Variable region (Fab) ✓**
 - Hinge region
 - Fc region
- Which cells mediate antibody-dependent cellular cytotoxicity (ADCC)?
 - Neutrophils
 - Eosinophils
 - NK cells ✓**
 - Basophils
- Which pathway of complement activation is triggered by antigen-antibody complexes?
 - Classical pathway ✓**
 - Alternative pathway
 - Lectin pathway
 - Properdin pathway
- C3 convertase in the classical pathway is:
 - C4b2a ✓**
 - C3bBb
 - C1qrs
 - C5b67
- Which component of complement is an anaphylatoxin?
 - C3a ✓**
 - C3b
 - C5b
 - C9
- Clonal deletion of self-reactive lymphocytes occurs in:
 - Peripheral tissues
 - Bone marrow (for B cells) & Thymus (for T cells) ✓**
 - Spleen
 - Lymph nodes
- Which MHC molecule presents antigen to CD8+ T cells?
 - MHC I ✓**
 - MHC II
 - CD1
 - HLA-G
- Which cytokine is essential for the growth and differentiation of B cells?
 - IL-2
 - IL-4 ✓**
 - IL-10
 - IFN-γ
- Delayed type hypersensitivity is mediated by:
 - IgE
 - IgG
 - T helper 1 cells (Th1) ✓**
 - Mast cells
- The first line of defense in innate immunity is:
 - Antibodies
 - Skin and mucous membranes ✓**
 - T lymphocytes
 - Complement system
- Which receptor on macrophages recognizes bacterial LPS?
 - TLR4 ✓**
 - TLR2
 - TLR9
 - TLR7
- Which of the following is a regulatory cytokine?
 - IL-2
 - IFN-γ
 - IL-10 ✓**
 - TNF-α
- What is the main function of regulatory T cells (Treg)?
 - Produce antibodies
 - Suppress immune responses ✓**
 - Activate B cells
 - Induce inflammation
- Which immunoglobulin can cross the placenta?
 - IgM
 - IgG ✓**
 - IgA
 - IgE
- Splenic marginal zone B cells are involved in response to:
 - Protein antigens
 - Polysaccharide antigens ✓**
 - Viral antigens
 - Self antigens
- Which structure in the lymph node traps antigens from lymph?
 - Capsule
 - Subcapsular sinus ✓**
 - Medullary cords
 - Hilum
- Which is a feature of autoimmune diseases?
 - Loss of self-tolerance ✓**
 - Increased tolerance
 - Lack of B cells
 - Deficiency of complement
- Which cell type is the most efficient antigen-presenting cell (APC)?
 - B lymphocyte
 - Macrophage
 - Dendritic cell ✓**
 - Neutrophil
- Which of the following is NOT a type of hypersensitivity?
 - Type I
 - Type II
 - Type III
 - Type V ✓**



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- Which cell is the major source of antigen presentation to CD4+ T cells?
A. B lymphocyte B. **Dendritic cell** ✓
C. Neutrophil D. Mast cell
- Immunological memory is primarily mediated by:
A. Neutrophils B. Eosinophils
C. **Memory B cells and T cells** ✓ D. Basophils
- Which immunoglobulin is associated with allergic reactions?
A. IgA B. IgG
C. **IgE** ✓ D. IgM
- The complement component that forms the membrane attack complex is:
A. C3b B. **C5b** ✓
C. C4b D. C3a
- Which of the following is NOT a primary lymphoid organ?
A. Bone marrow B. Thymus
C. **Spleen** ✓ D. Bursa of Fabricius
- The main function of IgA is:
A. **Neutralization at mucosal surfaces** ✓
B. Activation of complement
C. Opsonization of bacteria
D. Immediate hypersensitivity
- Which cytokine is produced by Th1 cells?
A. IL-4 B. IL-5
C. **IFN-γ** ✓ D. IL-10
- Natural killer (NK) cells are part of:
A. **Innate immunity** ✓ B. Humoral immunity
C. Adaptive immunity D. Acquired immunity
- Which region of antibody binds to antigen?
A. Fc region B. **CDR (complementarity determining region)** ✓
C. Hinge region D. Constant region
- Which hypersensitivity reaction is mediated by immune complexes?
A. Type I B. Type II
C. **Type III** ✓ D. Type IV
- The cytokine that promotes B cell class switching to IgE is:
A. IFN-γ B. **IL-4** ✓
C. IL-12 D. TGF-β
- Which of the following is a phagocytic cell?
A. Erythrocyte B. Platelet
C. **Macrophage** ✓ D. Plasma cell
- The recognition of "self" is known as:
A. **Tolerance** ✓ B. Memory
C. Sensitization D. Immunodeficiency
- Which molecule provides co-stimulatory signal to T cells?
A. MHC II B. **CD28 binding to B7 (CD80/86)** ✓
C. CD16 D. IgG
- The first barrier of innate immunity is:
A. Lysozyme B. **Skin and mucous membranes** ✓
C. Complement system D. Interferons
- Which immunoglobulin is the first to appear in primary immune response?
A. IgA B. IgG
C. **IgM** ✓ D. IgE
- The site where B cells mature is:
A. Thymus B. **Bone marrow** ✓
C. Spleen D. Peyer's patch
- Which of the following is a regulatory T cell marker?
A. CD8 B. CD4
C. **CD25 (IL-2Rα)** ✓ D. CD16
- Antigens that induce antibody response without T cell help are called:
A. T-dependent antigens
B. **T-independent antigens** ✓
C. Autoantigens
D. Neoantigens
- Which immunoglobulin can fix complement most efficiently?
A. **IgM** ✓ B. IgA
C. IgE D. IgD



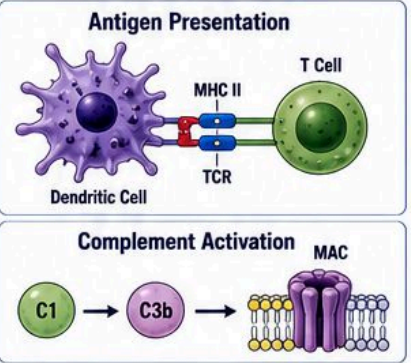
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SET 4



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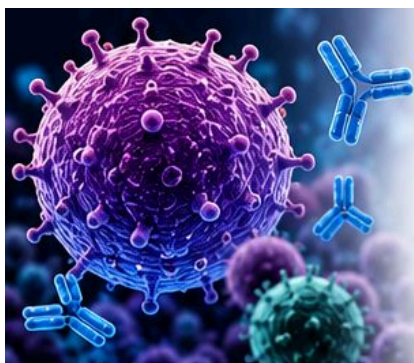
— VU PIONEERS OF ZOOLOGY —

- Which cytokine is primarily responsible for differentiation of naive CD4⁺ T cells into Th17 cells?
A. IL-2 B. IL-4
C. TGF- β D. IL-6 ✓
- Interleukin-2 (IL-2) is mainly produced by:
A. B lymphocytes B. Macrophages
C. Helper T cells (Th1) ✓ D. Neutrophils
- Which of the following is a potent pro-inflammatory cytokine?
A. IL-10 B. TGF- β
C. IL-12 ✓ D. IL-4
- Which is the main function of Interferon- γ (IFN- γ)?
A. Stimulates B cell proliferation
B. Activates macrophages ✓
C. Promotes eosinophil activation
D. Induces IgE production
- Which hypersensitivity reaction is caused by immune complex deposition?
A. Type I B. Type II
C. Type III ✓ D. Type IV
- In Type II hypersensitivity, antibodies are directed against:
A. Soluble antigens
B. Cell surface or extracellular matrix antigens ✓
C. Immune complexes
D. Transplanted tissues
- The Arthus reaction is an example of:
A. Type I hypersensitivity B. Type II hypersensitivity
C. Type III hypersensitivity ✓ D. Type IV hypersensitivity
- Which cell is mainly responsible for immediate (Type I) hypersensitivity?
A. Mast cell ✓ B. Eosinophil
C. Macrophage D. NK cell
- Which of the following is NOT a feature of Type I hypersensitivity?
A. IgE mediated B. Immediate response
C. Complement activation ✓ D. Release of histamine
- Which molecule present on antigen-presenting cells engages with CD28 on T cells for co-stimulation?
A. MHC II B. CD80/CD86 (B7 molecules) ✓
C. CD40 D. ICAM-1
- Which MHC class molecule presents antigen to CD8⁺ T cells?
A. MHC I ✓ B. MHC II
C. MHC III D. CD1d
- Which complement pathway is activated by lectin binding to carbohydrates on pathogens?
A. Classical pathway B. Alternative pathway
C. Lectin pathway ✓ D. Properdin pathway
- C3 convertase of the classical pathway is:
A. C3bBb B. C4b2a ✓
C. C4bBb3b D. C3b2a
- The terminal complement complex (Membrane Attack Complex) is formed by:
A. C3b, C5b B. C5b, C6, C7, C8, C9 ✓
C. C1, C2, C3 D. C4b, C6, C8, C9
- Which complement component is an important opsonin?
A. C1q B. C3b ✓
C. C9 D. C5a
- Which cytokine inhibits both Th1 and Th2 responses?
A. IL-2 B. IL-4
C. IL-10 ✓ D. IFN- γ
- Which of the following is a live attenuated vaccine?
A. BCG vaccine ✓ B. Tetanus toxoid
C. Inactivated polio vaccine D. Hepatitis B vaccine
- Which is an adjuvant used in many vaccines to enhance immune response?
A. Formalin B. Alum (Aluminum hydroxide) ✓
C. Penicillin D. Heparin
- Which type of immunity is transferred through breast milk?
A. Active natural B. Passive natural ✓
C. Active artificial D. Passive artificial
- Which of the following is an example of autoimmune disease?
A. Rheumatoid arthritis ✓ B. AIDS
C. Measles D. Tuberculosis



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SET 5

Hypersensitivity Types

Type I (Immediate)	Type II (Cytotoxic)
Type III (Immune Complex)	Type IV (Delayed)



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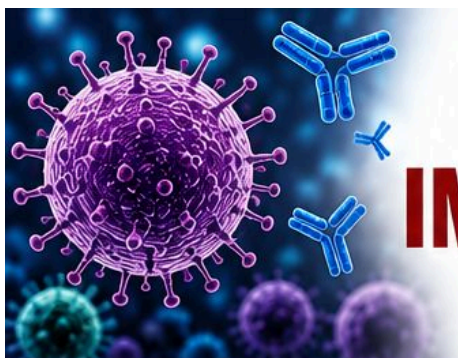
VU PIONEERS OF ZOOLOGY

- Which of the following is the primary site of T cell maturation?
 - Bone marrow
 - Thymus ✓**
 - Spleen
 - Peyer's patches
- Which molecule is essential for antigen processing and presentation on MHC class II?
 - TAP
 - Proteasome
 - Invariant chain (Ii) ✓**
 - CD28
- Which of the following is a Type IV (hypersensitivity) reaction?
 - Anaphylaxis
 - Serum sickness
 - Contact dermatitis ✓**
 - Hemolytic anemia
- Which cytokine is produced by Th2 cells that promotes IgE class switching?
 - IFN-γ
 - IL-2
 - IL-4 ✓**
 - IL-12
- Which complement component is involved in chemotaxis of neutrophils?
 - C1q
 - C3a ✓**
 - C5b
 - C9
- Which of the following is NOT a function of macrophages?
 - Phagocytosis
 - Antigen presentation
 - Antibody production ✓**
 - Cytokine secretion
- Which immunoglobulin is most effective in activating the classical complement pathway?
 - IgM ✓**
 - IgG
 - IgA
 - IgE
- The silhouette of a foreign antigen that is recognized by BCR or TCR is called:
 - Epitope ✓**
 - Paratope
 - Haplotype
 - Idiotype
- Which type of vaccine contains killed pathogens?
 - Live attenuated vaccine
 - Subunit vaccine
 - Inactivated (killed) vaccine ✓**
 - DNA vaccine
- Which molecule is important for the adhesion of leukocytes to vascular endothelium?
 - Selectins ✓**
 - Immunoglobulins
 - Complement proteins
 - Cytokines
- Which cytokine activates NK cells?
 - IL-4
 - IL-10
 - IL-12 ✓**
 - TGF-β
- The immune response that protects against extracellular bacteria is mainly:
 - Cell-mediated immunity
 - Humoral immunity ✓**
 - Innate immunity
 - Inflammatory response
- What is the main role of regulatory T cells (Treg)?
 - Antibody production
 - Suppression of autoimmunity ✓**
 - Activation of complement
 - Killing infected cells
- Which of the following is an example of a superantigen?
 - Tetanus toxoid
 - Staphylococcal enterotoxin ✓**
 - Influenza hemagglutinin
 - Hepatitis B surface antigen
- Which molecule is expressed on all nucleated cells and presents antigen to CD8+ T cells?
 - MHC class I ✓**
 - MHC class II
 - B7 molecule
 - ICAM-1
- Which of the following is a characteristic of immune memory?
 - Immediate response
 - Long-lasting and faster response ✓**
 - Non-specific
 - No involvement of lymphocytes
- Which organ stores mature B lymphocytes?
 - Bone marrow ✓**
 - Thymus
 - Spleen
 - Liver
- Which antibody is the predominant one in mucosal secretions?
 - IgG
 - IgA ✓**
 - IgM
 - IgE
- Which is the hallmark of autoimmune disease?
 - Response against self antigens ✓**
 - Response against foreign pathogens
 - Overproduction of antibodies
 - Deficiency of complement
- Which immunodeficiency is characterized by low IgG and IgA with normal or increased IgM?
 - SCID
 - X-linked agammaglobulinemia
 - Common variable immunodeficiency
 - Hyper IgM syndrome ✓**



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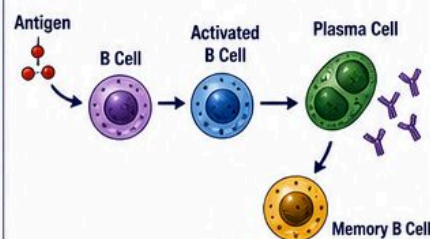




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SET 6

B Cell Activation & Differentiation



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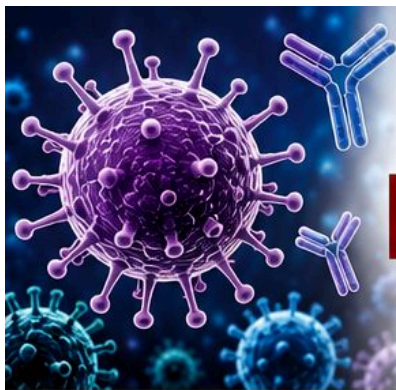
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- Which cell acts as the most potent professional antigen-presenting cell (APC)?
A. B lymphocyte B. Macrophage
C. Dendritic cell ✓ D. Neutrophil
- Which co-stimulatory signal is required along with antigen recognition for full T cell activation?
A. CD28 – B7 (CD80/CD86) interaction ✓
B. CD40 – CD40L interaction
C. MHC – TCR interaction
D. Cytokine – receptor interaction
- Which region of an antibody binds to the Fc receptor on phagocytes?
A. Fab region **B. Fc region ✓**
C. Variable region D. Hypervariable region
- Which of the following is a natural antibody present in the blood?
A. IgG B. IgA
C. IgM ✓ D. IgE
- Which complement component is the rate-limiting factor of the alternative pathway?
A. C1q **B. Factor B ✓**
C. C3b D. C5
- Which cytokine is mainly produced by macrophages and acts on T cells?
A. IL-2 **B. IFN-γ ✓**
C. IL-4 D. IL-10
- Which hypersensitivity reaction is associated with Arthus reaction?
A. Type I B. Type II
C. Type III ✓ D. Type IV
- Which of the following is a characteristic feature of autoimmunity?
A. Tolerance to self antigens **B. Loss of self tolerance ✓**
C. Increased complement activity D. High level of IgM
- Which vaccine is prepared from inactivated (killed) microorganisms?
A. BCG vaccine **B. DPT vaccine ✓**
C. OPV vaccine D. MMR vaccine
- Which cytokine promotes differentiation of naïve CD4⁺ T cells into Th1 cells?
A. IL-4 B. IL-10 **C. IFN-γ ✓**
- Which molecule is responsible for clonal deletion of self-reactive lymphocytes?
A. Interleukin-2 B. **Fas – FasL interaction ✓**
C. CTLA-4 D. CD28
- Which organ is the primary site for B cell maturation in humans?
A. Thymus B. Spleen
C. Bone marrow ✓ D. Lymph node
- Which of the following is a Th2 cytokine?
A. IFN-γ B. IL-2
C. IL-4 ✓ D. IL-12
- Which cell is the main source of IL-17?
A. Th1 cell **B. Th17 cell ✓**
C. Th2 cell D. Treg cell
- Which immunoglobulin is transferred across the placenta?
A. IgA B. IgM **C. IgG ✓** D. IgE
- Which complement pathway is activated spontaneously?
A. Classical pathway **B. Alternative pathway ✓**
C. Lectin pathway D. None of these
- Which molecule is expressed on activated T cells to help B cells?
A. CD3 **B. CD40L (CD154) ✓**
C. CD28 D. MHC I
- Which of the following is an example of Type II hypersensitivity?
A. Rheumatoid arthritis **B. Goodpasture syndrome ✓**
C. Serum sickness D. Contact dermatitis
- Which of the following is a passive artificial immunity?
A. Vaccination **B. Injection of antitoxin/antibody ✓**
C. Natural infection D. Breast feeding
- Which immunodeficiency is caused by defect in B cells leading to absence of all Ig classes?
A. HIV infection B. SCID
C. X-linked agammaglobulinemia ✓
D. DiGeorge syndrome



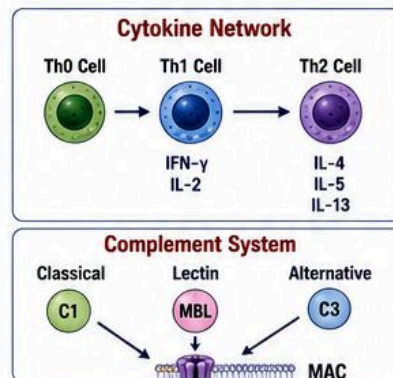
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SET 7



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- Which cell type is responsible for antibody secretion in the humoral immune response?
 - Helper T cell
 - Cytotoxic T cell
 - Plasma cell ✓**
 - Macrophage
- The "double positive" stage of T cell development (CD4+CD8+) occurs in:
 - Bone marrow
 - Thymic cortex
 - Thymic medulla ✓**
 - Spleen
- Which cytokine is essential for the growth and differentiation of B cells?
 - IL-2
 - IFN-γ
 - IL-4 ✓**
 - TNF-α
- Which immunoglobulin is most effective in opsonization?
 - IgA
 - IgG ✓**
 - IgM
 - IgE
- Which receptor on NK cells recognizes downregulated MHC class I molecules?
 - TCR
 - Fc receptor
 - KIR (Killer-cell immunoglobulin-like receptor) ✓**
 - CD28
- Which of the following is a characteristic of central tolerance?
 - Anergy
 - Clonal deletion ✓**
 - Immune deviation
 - Opsonization
- Which cytokine is produced by Th1 cells and activates macrophages?
 - IL-4
 - IL-10
 - IFN-γ ✓**
 - IL-5
- The formation of immune complex disease is an example of:
 - Type I hypersensitivity
 - Type II hypersensitivity
 - Type III hypersensitivity ✓**
 - Type IV hypersensitivity
- Which molecule is required for antigen processing in the endoplasmic reticulum?
 - Tapasin ✓**
 - MBL
 - C1q
 - CD28
- Which of the following is NOT a function of complement system?
 - Opsonization
 - Inflammation
 - Membrane attack
 - Antibody production ✓**
- Which complement component is an anaphylatoxin and causes mast cell degranulation?
 - C3a ✓**
 - C5b
 - C7
 - C9
- Which immunoglobulin is predominantly found in secretions like saliva, tears and colostrum?
 - IgA ✓**
 - IgG
 - IgM
 - IgE
- Which molecule is expressed on antigen-presenting cells and provides co-stimulatory signal to T cells?
 - B7 (CD80/CD86) ✓**
 - CD3
 - MHC II
 - FcR
- Which organ is the major site of immune response against blood-borne antigens?
 - Thymus
 - Bone marrow
 - Spleen ✓**
 - Lymph node
- Which cytokine promotes Th17 cell differentiation?
 - IL-4
 - IL-10
 - TGF-β + IL-6 ✓**
 - IFN-γ
- Which of the following is an example of a recombinant vaccine?
 - Hepatitis B vaccine ✓**
 - BCG vaccine
 - OPV
 - DPT vaccine
- Which hypersensitivity reaction is mediated by T cells rather than antibodies?
 - Type I
 - Type II
 - Type IV ✓**
 - Type III
- Which molecule binds to mannose residues on pathogens and activates the lectin pathway?
 - C1q
 - MBL ✓**
 - Factor B
 - Properdin
- Which of the following is a regulatory cytokine?
 - IL-2
 - IFN-γ
 - IL-10 ✓**
 - IL-6
- Which immunodeficiency is caused by defect in T cell development due to DiGeorge syndrome?
 - X-linked agammaglobulinemia
 - Wiskott-Aldrich syndrome
 - Severe combined immunodeficiency (SCID)
 - DiGeorge syndrome ✓**

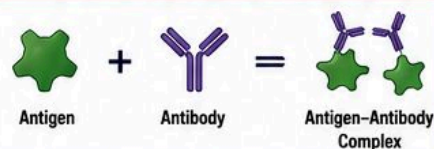


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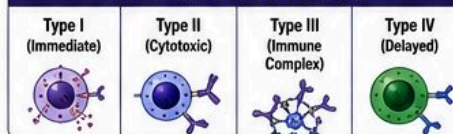


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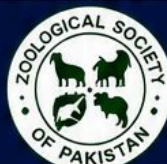
ANTIGEN-ANTIBODY REACTION



HYPERSENSITIVITY TYPES



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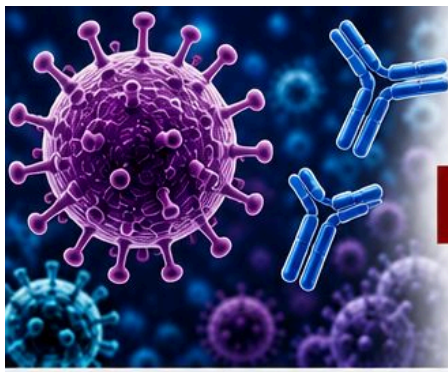
VU PIONEERS OF ZOOLOGY

- Which immunoglobulin is the first to be produced in primary immune response?
A. IgG B. IgA
C. IgM ✓ D. IgE
- Which complement component is known as the "central component" of complement system?
A. C1q B. **C3 ✓**
C. C5 D. C9
- Which cells are professional antigen-presenting cells (APCs)?
A. Neutrophils and Eosinophils
B. Dendritic cells, Macrophages and B cells ✓
C. NK cells and Mast cells
D. T cells and Plasma cells
- Which cytokine is mainly produced by Th1 cells and activates macrophages?
A. IL-4 B. IL-10
C. IFN-γ ✓ D. TGF-β
- Which of the following is an example of Type I (immediate) hypersensitivity?
A. Serum sickness B. **Allergy (e.g. hay fever) ✓**
C. Arthus reaction D. Contact dermatitis
- Which molecule is essential for antigen processing in the cytosol for MHC class I presentation?
A. TAP B. **Proteasome ✓**
C. Tapasin D. Calnexin
- Which cell type is mainly responsible for antibody production?
A. T helper cell B. **B lymphocyte ✓**
C. Macrophage D. NK cell
- Which immunoglobulin does NOT efficiently activate complement?
A. IgM B. IgG C. **IgA ✓**
- Which vaccine uses weakened but living microorganisms?
A. Salk vaccine B. Subunit vaccine
C. Live attenuated vaccine ✓ D. Toxoid vaccine
- Which cytokine is known as the "master regulator" of immune responses?
A. IL-2 B. IFN-γ
C. IL-10 D. **TGF-β ✓**
- Which complement pathway is triggered by antigen-antibody complexes?
A. Alternative pathway B. Lectin pathway
C. **Classical pathway ✓** D. Properdin pathway
- Which molecule prevents activation of self-reactive lymphocytes?
A. **Anergy ✓** B. Clonal expansion
C. Inflammation D. Opsonization
- Which receptor on B cells recognizes native (unprocessed) antigen?
A. TCR B. **BCR (Surface Ig) ✓**
C. FcR D. MHC II
- Which organ is the major site for maturation of T lymphocytes?
A. Bone marrow B. Spleen
C. Thymus ✓ D. Lymph node
- Which cytokine is produced by Treg cells and suppresses immune responses?
A. IL-2 B. IL-4
C. TGF-β D. **IL-10 ✓**
- Immune complexes deposit in blood vessels and activate complement causing:
A. Type I hypersensitivity B. Type II hypersensitivity
C. Type III hypersensitivity ✓ D. Type IV hypersensitivity
- Which molecule is expressed on all nucleated cells and presents antigen to CD8+ T cells?
A. **MHC class I ✓** B. MHC class II
C. CD28 D. ICAM-1
- Which of the following is an example of autoimmune disease?
A. **Rheumatoid arthritis ✓** B. Tuberculosis
C. Measles D. Tetanus
- Which immunodeficiency is caused by defect in neutrophil chemotaxis?
A. SCID B. DiGeorge syndrome
C. Chronic granulomatous disease ✓
D. X-linked agammaglobulinemia
- Which molecule cross-links and activates mast cells in allergy?
A. IgG B. IgA C. **IgE ✓** D. IgM



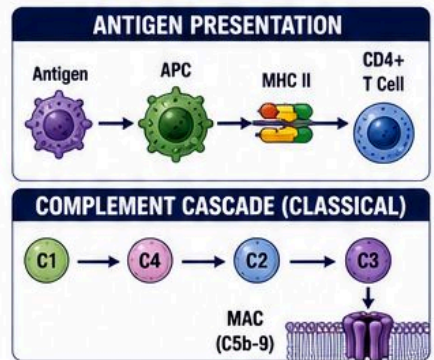
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BT302 IMMUNOLOGY

SET 9



Composed & Compiled by
Imtiaz Ahmad (Albaloshi)



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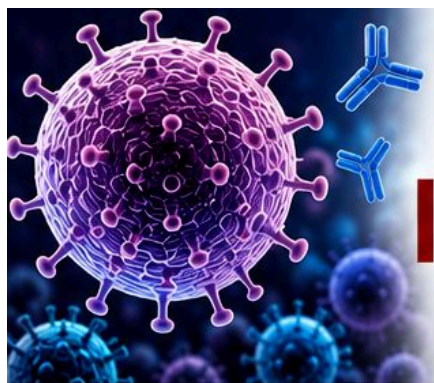
VU PIONEERS OF ZOOLOGY

- Which is the primary function of MHC class I molecules?
 - Present antigen to CD4+ T cells
 - Present antigen to CD8+ T cells ✓**
 - Activate B cells
 - Bind antibodies
- Which cytokine inhibits the production of inflammatory cytokines?
 - IFN- γ
 - IL-1
 - IL-10 ✓**
 - TNF- α
- Which complement component acts as an opsonin?
 - C3a
 - C3b ✓**
 - C5a
 - C9
- Which cell type is the main source of interleukin-2 (IL-2)?
 - Macrophage
 - B lymphocyte
 - Activated T helper (Th1) cell ✓**
 - Mast cell
- Which hypersensitivity reaction is caused by deposition of immune complexes in tissues?
 - Type I
 - Type II
 - Type III ✓**
 - Type IV
- Which molecule is essential for the activation of the classical complement pathway?
 - Properdin
 - C1q ✓**
 - Factor D
 - C3b
- Which immunoglobulin is the first to appear in response to antigen?
 - IgG
 - IgA
 - IgM ✓**
 - IgE
- Which organ contains the highest concentration of lymphocytes in the body?
 - Liver
 - Spleen ✓**
 - Lungs
 - Kidneys
- Which cytokine is produced by Th2 cells and promotes B cell class switching to IgE?
 - IFN- γ
 - IL-2
 - IL-4 ✓**
 - IL-12
- Which vaccine is prepared from purified polysaccharide antigens?
 - Conjugate vaccine
 - Subunit (polysaccharide) vaccine ✓**
 - Live attenuated vaccine
 - Toxoid vaccine
- Which molecule is responsible for the adhesion of lymphocytes to high endothelial venules?
 - Selectins ✓**
 - Integrins
 - Immunoglobulins
 - Complement proteins
- Which of the following is NOT a function of T helper cells?
 - Cytokine secretion
 - B cell help
 - Direct antibody secretion ✓**
 - Activation of macrophages
- Which cell is primarily involved in immediate release of histamine during Type I hypersensitivity?
 - Eosinophil
 - Mast cell ✓**
 - Neutrophil
 - Basophil
- Which MHC class molecule presents antigen from extracellular pathogens?
 - MHC class I
 - MHC class II ✓**
 - MHC class III
 - CD1d
- Which complement fragment is a potent chemoattractant for neutrophils?
 - C3a
 - C5a ✓**
 - C1q
 - C9
- Which immunodeficiency is characterized by recurrent bacterial infections due to defective phagocyte killing?
 - X-linked agammaglobulinemia
 - Chronic granulomatous disease ✓**
 - SCID
 - DiGeorge syndrome
- Which cytokine promotes differentiation of Th17 cells?
 - IL-4
 - IL-6 + TGF- β ✓**
 - IFN- γ
 - IL-2
- Which antibody crosses the placenta and provides natural passive immunity to the fetus?
 - IgA
 - IgM
 - IgG ✓**
 - IgE
- Which cell type is responsible for antigen presentation via MHC class II to naive T cells?
 - Dendritic cell ✓**
 - RBC
 - Neutrophil
 - Platelet
- Which hypersensitivity is mediated by sensitized T cells and macrophage activation?
 - Type I
 - Type II
 - Type III
 - Type IV ✓**

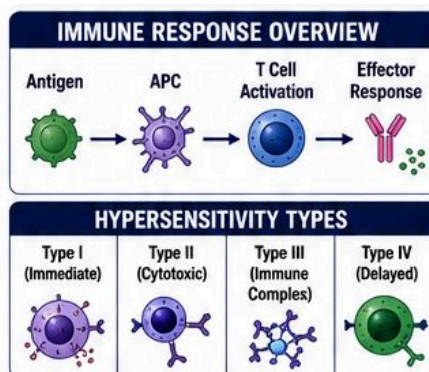


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BT302 IMMUNOLOGY SET 10



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- Which molecule on APCs binds to CD28 on T cells to provide co-stimulatory signal?
A. CD40 B. **B7 (CD80/CD86) ✓**
C. MHC II D. ICAM-1
- Which type of immunity provides immediate protection but is short-lived?
A. Active natural B. Passive natural
C. Active artificial D. **Passive artificial ✓**
- Which complement component is responsible for forming the Membrane Attack Complex (MAC)?
A. C3b B. C5a
C. **C9 ✓** D. C1q
- Which cytokine is essential for the survival and proliferation of B cells?
A. IL-2 B. IL-12
C. **IL-4 ✓** D. IFN-γ
- Which cell type produces perforin and granzymes to kill infected cells?
A. B lymphocyte B. **CD8+ T lymphocyte (CTL) ✓**
C. Macrophage D. Mast cell
- Which receptor on B cells binds to the Fc portion of IgG and internalizes immune complexes?
A. TCR B. CD4
C. **FcγR (CD16/CD32) ✓** D. MHC II
- Which immunoglobulin is most abundant in mucosal secretions?
A. IgG B. **IgA ✓** C. IgM D. IgE
- Which disease is caused by deficiency of ADA enzyme leading to accumulation of toxic metabolites?
A. X-linked agammaglobulinemia
B. **Severe combined immunodeficiency (SCID) ✓**
C. Chronic granulomatous disease
D. DiGeorge syndrome
- Which pathway of complement activation is triggered by microbial polysaccharides and lipopolysaccharides?
A. Classical B. Lectin
C. **Alternative ✓** D. Properdin
- Which cytokine is mainly involved in neutrophil chemotaxis and activation?
A. IL-4 B. IFN-γ
C. **IL-8 (CXCL8) ✓** D. IL-10
- Which molecule on endothelial cells interacts with L-selectin on leukocytes to mediate initial rolling?
A. ICAM-1 B. VCAM-1
C. P-selectin D. **E-selectin ✓**
- Which structure of antibody binds to antigen?
A. Fc region B. **Fab region ✓**
C. Hinge region D. Constant region
- Which cytokine is produced by macrophages and promotes Th1 differentiation?
A. IL-4 B. IL-10
C. **IL-12 ✓** D. TGF-β
- Which of the following is an example of autoimmune disease?
A. **Rheumatoid arthritis ✓** B. Tuberculosis
C. Tetanus D. Typhoid fever
- Which MHC class molecule is expressed on all nucleated cells?
A. **MHC class I ✓** B. MHC class II
C. MHC class III D. CD1d
- Which cell presents antigen to naive CD4+ T cells?
A. Neutrophil B. **Dendritic cell ✓**
C. Erythrocyte D. Platelet
- Which immunoglobulin is the main antibody in secondary immune response?
A. IgM B. IgA C. **IgG ✓** D. IgE
- Which hypersensitivity reaction is mediated by cytotoxic T cells?
A. Type I B. Type II C. Type III D. **Type IV ✓**
- Which cytokine inhibits macrophage activation and promotes anti-inflammatory responses?
A. IFN-γ B. IL-12 C. **IL-10 ✓** D. IL-2
- Which immunodeficiency is caused by defect in NADPH oxidase enzyme of phagocytes?
A. SCID
B. **Chronic granulomatous disease ✓**
C. DiGeorge syndrome
D. X-linked agammaglobulinemia



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